

Benchmarking Quantum Algorithms for the Burgers' Equation based on the Lattice Boltzmann Method and the Cole–Hopf Linearization

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ABSTRACT

Abstract—Quantum computing for solving partial differential equations at utility scale, either independently or as an assist to classical solvers, is an emerging field whose utility has not yet been fully demonstrated. Existing algorithms which suggest an advantage may not obtain practical utility until the advent of fault-tolerant quantum computing. Until then, studies can begin to compare implementations using scalable cases and identify usefulness in a future engineering toolbox. Given the rapidly changing quantum computing landscape, some care needs to be taken in setting up and capturing those experiments and their environmental metadata for long-term comparison.

Toward that end, we implement and compare two distinct approaches to solve the 1D viscous Burgers' equation: (1) a variation on the quantum lattice Boltzmann method which attempts to address the inherent nonlinearity, and (2) a case-applicable linearization proposed by Cole and Hopf followed by a quantum diffusion step. We report the errors in the quantum solutions, quantum resource utilization, and clock times and discuss suggestions for future comparisons.

Index Terms—quantum computing, partial differential equations, Burgers' equation, Cole–Hopf transformation, lattice Boltzmann method, CFD, NISQ, FTQC

I. INTRODUCTION

The use of quantum computing as an alternative to or accelerator for solving partial differential equations (PDEs), for example in computational fluid dynamics (CFD) applications, has garnered both attention and skepticism in this current noisy intermediate-scale quantum (NISQ) era. For example, the problem of solving nonlinear systems on an inherently linear quantum computer is a motivating stimulus for ongoing research.

Even admitting no plausible utility until the advent of stable and resource-rich fault-tolerant quantum computing (FTQC), a gap remains in understanding applicability - utility first, then perhaps advantage. Quantum approaches are varied, in both the portion of the PDE solver or pre-processor [1] being addressed, as well as the algorithm being employed. While there may be leading candidates, for example the well-studied HHL algorithm [2], they come with tradeoffs which

impact their utility to specific cases. Understanding which cases are practically amenable to which algorithms is an area of continued research, further complicated by the rapid change and short shelf life of quantum software and hardware layers. Thus, further novel and deep algorithmic study is needed, as well as a broad survey applying engineered implementations of theoretical algorithms to concrete cases, e.g. [3]. In a broad survey, fine-grained data and metadata collection is needed to provide an objective view of results, for reusability and comparability over time where conditions to exactly reproduce may not exist [4].

It is toward this end that we conduct a comparative study of two distinct quantum approaches to solve a PDE of practical interest. Providing implementations where needed (relevant codes and results in [5]), we focus on understanding the algorithms' relative applicability and specifics of their software engineering. In selecting the algorithms for this study, several constraints which govern practical utility are enforced:

- C-1** The algorithm must be applicable to the case and provide a full state vector solution, not just statistics.
- C-2** The solver may not maintain a shadow classical solution to guide, normalize, or otherwise steer the quantum solution.
- C-3** Classical pre- and post-processing is allowed. Pseudo-time stepping, a typical aspect of classical solvers, is allowed, where each step is solved by an iteration of the code.
- C-4** The solver must return a signed solution where needed. For some algorithms, this comes at an additional cost.
- C-5** The implementation must be hardware-ready. This usually means a shots-based implementation, even when a simulator is used for testing.
- C-6** The code may bring its best configuration suitable for the case being solved - tunable parameters may be set for best case-specific results.
- C-7** The code may use no more than $C \times$ the number of identified qubits, e.g. for ancilla, during processing. This allows algorithms to trade off circuit width and depth options.
- C-8** Rich metadata must be captured with the result.

Metadata includes information about the case, the code (e.g. versions of dependent libraries), real or simulated hardware

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specs with coupling map and calibration data, and fine-grained runtime information. While we capture runtimes and can compare competing implementations, we do not endeavor to demonstrate runtime advantage over classical in this NISQ era. Nor do we intend to demonstrate accuracy or scalability relative to classical - here the comparisons are between competing quantum algorithms, where the classical solution serves as a reference or ground truth - if an algorithm can demonstrate results within a reasonable tolerance of the classical solution, a pathway to utility might exist in an FTQC era. Similarly, for scaling of the problem size - a problem 20% of a typical classical problem might be considered notably large for industrial application of quantum computing at this time. Also similar would be the utilization of a lower complexity computational model rather than the full Navier-Stokes equations.

Toward these goals, the case selected for comparison is the viscous one-dimensional (1D) Burgers' equation, a nonlinear PDE that serves as a model for advection-diffusion:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad (1)$$

where $u(x, t)$ is the velocity field and ν is the kinematic viscosity.

At this point in the research we avoid the rotational physics which makes Navier-Stokes hard, we omit the source term, and we drive the codes with cases which are expected to be within or on the edge of their regimes of utility. The approach and implementation of the two algorithms being compared for solving the 1D Burgers' equation are now presented.

II. QUANTUM LATTICE BOLTZMANN APPROACH

The well-studied lattice Boltzmann method (LBM) [6] has been implemented both classically and with emergent quantum computing techniques [7]. The flow field is segmented into a grid, 1D in our case, uniformly. At every cell, fictitious particles representative of the velocity are tracked during collisions for a fixed pseudo-time period in what represents the core nonlinear aspect of the Burgers' equation. Next, pseudo-particles are quantitatively streamed to the nearest neighbor grid element and the process repeats for the next time step. In our implementation, the collision portion is implemented in a quantum circuit, however to avoid unmanageably deep circuits the streaming is handled classically, albeit at a loss of precision due to decoupling. This is akin to the measure-and-reprepare pattern which we will employ for time stepping in the competing algorithm, discussed in Section III, and is permissible in a fair comparison given constraint C-3 above.

A typical QLBM implementation might suffer from the inability to discern the sign of the solution field elements without additional costly circuit constructions such as the Hadamard test [8]. This is in addition to the issues of solving a nonlinear problem with this algorithm on a linear quantum computer. Many quantum PDE solvers approach this problem via some form of linearization. The Quantum Algorithm for Lattice Boltzmann (QALB) [9] attempts to address the nonlinear problem directly. In QALB, the quasi-particle densities

are amplitude encoded as harmonic oscillators, allowing the code to take as argument the number of qubits to use per density register, with 3 per density register (9 total for D1Q3 LBM scheme) being used in this study in a tradeoff of accuracy to circuit depth and width. Here, the nonlinearity becomes an operator, and the QALB algorithm then evolves the initial state through a BGK collision simulation using a Hermitized collision Hamiltonian. The measurement is taken from the shots and classically processed, followed by the classical streaming procedure, before the loop repeats until the specified end of run. Time stepping given the split between quantum collision and classical streaming is fixed at one step - to integrate quantum streaming would further deepen the circuits, already at a concerning depth. This differs from the competing algorithm below, which internally and coherently can evolve through interim sub-steps. In this quantum LBM variation the sign is intrinsic to the readout - a negative quasi-particle population yields a negative expectation value, thus no additional sign discernment step is needed. Due to incomplete implementation of the full QALB proposal, we will refer our implementation in this paper using the generic term *QLBM*.

LBM itself is an approximation as is QLBM. Using the Forward Time Central Space (FTCS) as also an approximate classical reference in the benchmark for L2 error suggests multiple error gaps to track for comparison, again requiring the need for rich data capture and preservation.

III. COLE-HOPF LINEARIZATION APPROACH

An alternative approach to the problem of the nonlinear term in the Burgers' equation is to sidestep it by linearization, and given the variety of choices, we are motivated to select a linearization well suited for the 1D Burgers case. The classical Cole-Hopf transformation [10]–[13] maps the nonlinear viscous Burgers' equation onto the linear heat equation $\phi_t = \nu \phi_{xx}$, where the nonlinearity is removed exactly, reducing to linear diffusion of ϕ . This state is then prepared into the quantum circuit with amplitude encoding, or optionally with a matrix product state encoding (MPS) with bond dimension providing tunable compression [14], [15], consistent with viewing the implementation as a reduced order model. Given that the Laplace operator is non-unitary, a block encoding is used. Working in the Laplacian eigenbasis, controlled rotations perform the heat equation dissipation for that time-step followed by a reverse transformation of the eigenbasis and a measurement.

What we will call the *CH* approach could evolve the system to the end of the desired time period, at a cost of incurring the error introduced by deep circuits, manifested by geometrically decaying sample retention and wasted shots. Or, given we are evolving the linear heat equation, reset the initial condition periodically with a measure-and-reprepare iterative approach, accepting error at the time-step interfaces. Having conceded any objective of demonstrating quantum supremacy over classical based on runtime, we accept the runtime costs of these repeated measurements and preparations in order to compare the approach to other quantum-centric

methods for a future FTQC era. The code provides a time-step size, which can be aligned with the same for the QLBM implementation and the classical reference (the latter which may be interpolated). Within the time-step, sub-steps can be coherently processed. The sign of the solution is captured by the Cole–Hopf transformation, returned by the measurements and classical reverse Cole–Hopf transformation for a result requiring no further sign discernment. Prior to the reverse Cole–Hopf, a Fourier low-pass filter with tunable number of ϕ modes is used to smooth the signal, in accordance with constraint C-6 above. No attempt was made to use this smoothing to adaptively reduce the shots budget, and we defer those adaptations to potential future work.

IV. CASES & RESULTS

To begin to compare the performance of our named quantum-classical CH and QLBM codes in regions of mutual applicability as well as edge cases, a variety of runs were conducted of both Burgers’ equation solvers. All runs used the same sine wave initial condition and periodic boundary while varying the amplitude, the kinematic viscosity ν , and the parameters of the time stepping to some set percentage of the shock wave development. All cases exhibited some nonlinearity, but as this is not a stress test, the effect was at most moderate in strength. The selection of a sine wave means sign discernment is pertinent and required. The number of qubits, q , is provided as input, though the code may add ancilla up to the constrained limit. The grid resolution is set by q and in both compared implementations the grid is uniform.

Each code was allowed to set its own internal parameters in accordance with the constraints presented above, to trade circuit width and depth, to control internal time stepping, and here, admittedly, more study is needed to better understand those optimal settings. In Figure 1, for example, CH exhibits ill behavior at $q = 8$ and beyond, likely due to untuned parameters and the tradeoffs of time-step size versus circuit depth. Runs were conducted on IBM Aer simulator with shots to represent statistical noise. For CH, we also ran on real IBM hardware, which included the use of error mitigation techniques built into the execution platform, including twirled readout-error extinction (TREX) and Dynamical Decoupling (DD) [16].

Figure 1 denotes a comparison of the CH and QLBM solvers for a candidate case setting: $\nu = 0.03$, max amplitude $A = 0.3$, $cfl = 0.1$ (for setting the time-step), $q = [4, 6, 8]$, and time evolved to 30% of shock formation. While the total number of qubits required for CH increases with the problem size, QLBM can have a constant number of qubits for its workflow. CH is consistently wider (total qubits) but more shallow (max. depth) than QLBM. As a result, in regimes where CH is stable, CH is consistently faster: a $q = 6$ run of CH takes 3 sec versus 58 sec for QLBM. The accuracy of the solution in time shows that a smaller CH model produces results comparable to a larger QLBM.

We also compared the solution accuracy of the methods for a much more evolved state of the simulation — to a sharper

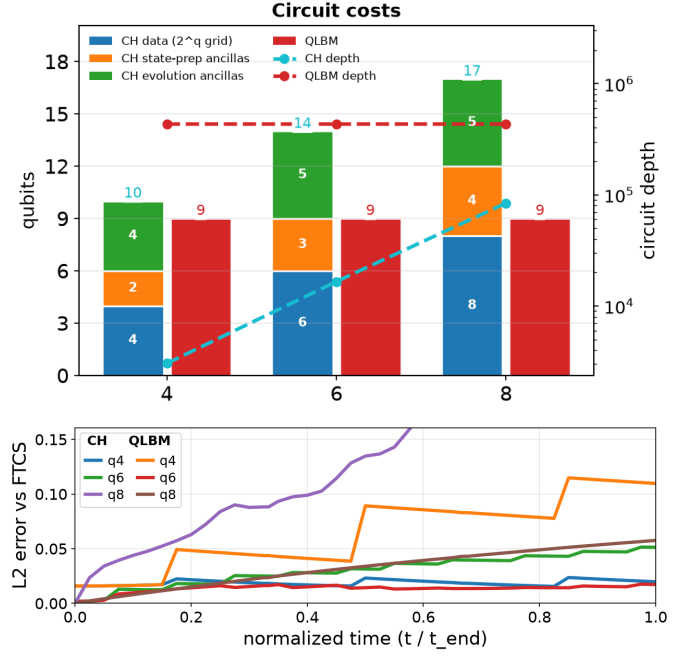


Fig. 1. Circuit statistics and solution fidelity comparison of QLBM and CH for a candidate case. Top: Qubits and circuit depth costs, showing CH qubits allocated for data, state prep, and evolution, versus those for QLBM as a function of problem size q . QLBM uses a constant number of qubits for D1Q3. Circuit depth is constant but deep for QLBM at these problem sizes, while CH grows with q . Bottom: L2 error versus FTCS over time using Aer simulator with 150k shots. CH is advantageous at small q , but at larger requires further algorithmic tuning.

50% ($\nu=0.025$, $A = 0.5$) and 90% ($\nu=0.03$, $A = 0.4$) shock formation. While CH offered higher accuracy for smaller q , QLBM improved with q , but had its own operating regimes (e.g. unstable at small ν) and circuit depth remained a concern. CH tunings to improve future runs with finer grid resolution include adapting the iteration size and the ϕ modes high-frequency shots noise filter.

Based on the above simulator results — of a small q problem with shallow circuits, amenable to the constraints of current NISQ devices — we run a candidate case ($\nu=0.08$, $A=0.3$, $q=3$, $cfl=0.3$, and time evolved to 40% of shock) using the CH method on various IBM devices chosen based on their performance (Heron R2 and R3) and qubit connectivity (Heron and the newer Nighthawk). We show the results (full state reconstruction) of only the first time-step in Figure 2, as the error subsequently compounds to noise. The Nighthawk performed best on the L2 metric compared to classical reference, while the runs on the Heron R2 and R3 showed no identifiable generational improvement. Admittedly these were limited tests, and while the results are promising to capture the overall nature of the wave, deeper understanding of the impact of the backend configuration, such as from the square lattice coupling on the Nighthawk backend, on the results and in quantifying uncertainty is a subject of future work. QLBM was not practical for real deployment at the required circuit depth.

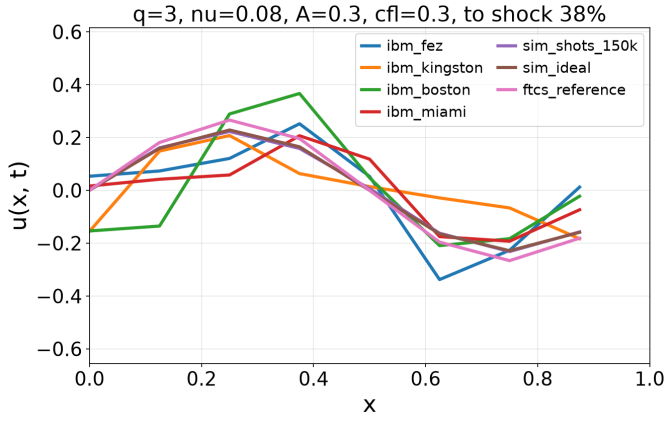


Fig. 2. CH runs of a small model size on contemporary IBM superconducting quantum computers, in order of maturity: Kingston (Heron R2), Fez (Heron R2), Boston (Heron R3), Miami (Nighthawk R1). Results of the state vector simulation (ideal) and with shots (150k) are also shown for comparison.

Comprehensive metadata about the run was captured for preservation, potentially enabling meaningful comparisons with future experiments. Time will tell if the proper data and provenance was preserved.

V. CONCLUSIONS & FUTURE WORK

CH may hold promise for certain cases in its regime of utility as an inexpensive estimation technique. QLBM accuracy improves with qubits, but with prohibitively deep circuits whereas CH can be coaxed into running on small quantum machines with longer time steps, balancing the reduced error of time-step length with the error due to circuit depth. While no specific conclusions can yet be drawn from these limited runs—in this comparison each algorithm demonstrated some strengths and weaknesses—it is clear further hands-on study is needed applying real quantum implementations to specific target cases.

Topics for future work include continuing these and similar *case × code* comparisons using the noted approach to metadata capture with open source and closed commercial codes on simulators and varied vendor backends. Improvements to the QLBM include extending to D1Q5, 2D, use of quantum streaming, use of Linear Combination of Unitaries (LCU), and intra-segment time stepping ($k > 1$). Improvements to the CH code include use of adaptive shots, other propagators, time-step optimizations, and tuning the ϕ modes smoothing. These activities can help identify a method’s ideal operating parameters in its regimes of utility, the engineering tradeoffs, and the potential dependency on hardware architectures including integration with the classical computers needed for a deployable solution. Such a body of understanding, if properly stewarded, would likely be useful to a practicing quantum CFD community.

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The authors acknowledge the use of modern code-assist tools based on Large Language Models (LLMs) and the code is made available as part of this publication [5]. However, no LLMs or tooling other than spellcheck was used to write the text of this paper, which remains the work of the authors.

REFERENCES

- [1] W. Aboumradi, D. Zhu, C. Giroto, F.-H. Rouet, J. Jojo, R. Lucas, J. Pathak, A. Kaushik, and M. Roetteler, “Accelerating large-scale linear algebra using variational quantum imaginary time evolution,” 2025. <https://arxiv.org/abs/2503.13128>
- [2] A. W. Harrow, A. Hassidim, and S. Lloyd, “Quantum algorithm for linear systems of equations,” *Physical Review Letters*, vol. 103, no. 15, p. 150502, 2009. <https://doi.org/10.1103/PhysRevLett.103.150502>
- [3] D. Jaksch, P. Givi, A. J. Daley, and T. Rung, “Variational quantum algorithms for computational fluid dynamics,” *AIAA Journal*, vol. 61, no. 5, pp. 1885–1894, 2023. <https://arxiv.org/abs/2209.04915>
- [4] M. D. Wilkinson, M. Dumontier, I. J. Aalbersberg, G. Appleton, M. Axton, A. Baak *et al.*, “The FAIR guiding principles for scientific data management and stewardship,” *Scientific Data*, vol. 3, p. 160018, 2016. <https://doi.org/10.1038/sdata.2016.18>
- [5] A. Gallo, “q8020-cfd-ch-ibm: Quantum CFD solver for the 1D Burgers equation using Cole–Hopf linearization vs LBM,” 2026. <https://github.com/Q8020-CFD/q8020-cfd-ch-ibm>
- [6] G. R. McNamara and G. Zanetti, “Use of the Boltzmann equation to simulate lattice-gas automata,” *Physical Review Letters*, vol. 61, no. 20, pp. 2332–2335, 1988. <https://doi.org/10.1103/PhysRevLett.61.2332>
- [7] A. Bastida-Zamora, L. Budinski, O. Kerppo, V. Lahtinen, O. Niemimäki, W. Steadman, R. Zamora-Zamora, P. Sagaut, V. Bohun, M. Koch-Janusz, and I. Lukin, “Quantum algorithm for the lattice boltzmann method with applications on real quantum devices,” 2026. <https://arxiv.org/abs/2603.02127>
- [8] D. Aharonov, V. Jones, and Z. Landau, “A polynomial quantum algorithm for approximating the Jones polynomial,” *arXiv preprint quant-ph/0511096*, 2005.
- [9] W. Itani, K. R. Sreenivasan, and S. Succi, “Quantum algorithm for lattice Boltzmann (QALB) simulation of incompressible fluids with a nonlinear collision term,” *Physics of Fluids*, vol. 36, no. 1, p. 017112, 2024, arXiv:2304.05915 [quant-ph]. <https://doi.org/10.1063/5.0176569>
- [10] J. D. Cole, “On a quasi-linear parabolic equation occurring in aerodynamics,” *Quarterly of Applied Mathematics*, vol. 9, no. 3, pp. 225–236, 1951. <https://doi.org/10.1090/qam/42889>
- [11] E. Hopf, “The partial differential equation $u_t + uu_x = \mu u_{xx}$,” *Communications on Pure and Applied Mathematics*, vol. 3, no. 3, pp. 201–230, 1950. <https://doi.org/10.1002/cpa.3160030302>
- [12] S. Jin and N. Liu, “Quantum algorithms for viscosity solutions to nonlinear Hamilton–Jacobi equations based on an entropy penalisation method,” 2025. <https://arxiv.org/abs/2512.07919>
- [13] F. Uchida, K. Miyamoto, S. Yamazaki, K. Fujisawa, and N. Yoshida, “Quantum simulation of Burgers turbulence: Nonlinear transformation and direct evaluation of statistical quantities,” 2024, authors’ code (Mathematica/Jupyter): https://github.com/fu-230/QC_Burgers. <https://arxiv.org/abs/2412.17206>
- [14] S.-J. Ran, “Encoding of matrix product states into quantum circuits of one- and two-qubit gates,” *Physical Review A*, vol. 101, no. 3, p. 032310, 2020. <https://doi.org/10.1103/PhysRevA.101.032310>
- [15] M. Gopalakrishnan Meena *et al.*, “A tensor network-based quantum algorithm for the nonlinear 1-d Burgers’ equation,” in *AIAA SciTech Forum*, 2026. <https://doi.org/10.2514/6.2026-1932>
- [16] A. Javadi-Abhari, M. Treinish, K. Krsulich, C. J. Wood, J. Lishman, J. Gacon, S. Martiel, P. D. Nation, L. S. Bishop, A. W. Cross, B. R. Johnson, and J. M. Gambetta, “Quantum computing with qiskit,” *arXiv preprint arXiv:2405.08810*, 2024. <https://arxiv.org/abs/2405.08810>